

The predecessor of RFID technology is considered to be a device that was invented in 1946 by Leon Theremin as an espionage tool for the Soviet Union. This was, in fact a covert listening device that transmitted radio waves with audio information.



An RFID tag used for electronic toll collection

The era of RFID can be first solidly traced back to a patent in 1973 by Mario Cardullo. This consisted of a passive transponder with memory. The device is passive but it can be made active by means of an interrogating signal. Cardullo himself foresaw the uses to which this new technology would be put, and he demonstrated it in 1971 to the New York Port Authority and other potential costumers as a means to collect tolls. The name RFID was first used in a patent that was granted in 1983 to Charles Walton.

An RFID device essentially consists of two parts: an integrated circuit and a receiving device. The integrated circuit consists of circuitry that stores the required information and also modulates and demodulates the RF (radio frequency) waves. The receiving device on the other hand receives and transmits the required signals.

RFID tags can be divided into three main groups: active, passive and battery assisted power (BAP). Active tags can transmit signals by themselves as they contain batteries. Passive tags, on the other hand, can transmit signals only in the presence of an external source. BAPs require an external source to transmit, but they also contain a battery and thus their signals are strong, providing great range.

The reason that RFID is so successful is the fact that the devices can be made really small. This makes them unobtrusive and easy to incorporate into pretty much anything. The smallest radio transponder in the world is the Hitachi μ -chip (or mu-chip). It measures only 0.4 mm by 0.4 mm – rather

The simplest example of an RFID system is the access cards used by companies to govern employee movement